Cantera vs NN | Test IC 04 | NRMSE (util.py)

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0108	0.0583	0.593	2.33	0.0953	0.832	0.0354	0.149	0.418	0	Complete
2 P-only DeepONet10	**10k**	8.74e-03	0.0113	0.165	0.217	0.0395	0.0155	2.22e-03	0.206	0.266	0	Complete
2 P-only DeepONet10	**long**	2.04e-03	1.92e-03	0.0153	0.0608	0.0293	0.0552	2.39e-03	0.0633	0.0794	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	9.87e-03	0.0137	0.196	0.258	0.0434	0.031	2.92e-03	0.155	0.2	0	Complete
3 ANN10 + P-only DeepONet10	**long**	7.36e-04	1.14e-03	0.0116	0.026	8.30e-03	0.0299	1.04e-03	0.0362	0.0393	0	Complete
4 T/P DeepONet10	**10k**	0.0189	0.0104	0.227	0.392	0.0858	0.218	9.30e-03	0.336	0.28	0	Complete
4 T/P DeepONet10	**long**	0.154	0.0899	1.66	5.28	0.0687	3.15	0.111	0.543	105	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.016	8.47e-03	0.186	0.267	0.0804	0.21	8.43e-03	0.233	0.181	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.111	0.0576	0.958	2.81	0.0298	3.18	0.105	0.116	0.589	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.0714	0.0485	0.943	0.811	0.196	0.973	0.0223	2.13	23.7	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.116	0.0915	1.74	1.93	0.0987	0.735	0.0284	0.368	2.06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	4.70e-03	3.41e-03	0.059	0.0529	0.0354	0.0625	2.65e-03	0.16	0.106	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.101	0.0863	1.57	1.33	0.0648	0.735	0.0245	0.289	1.82	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.